Pharma truth Finder

Counterfeit Medicine Detection & Awareness Assistant

Newton School of Technology

Problem Statement :

An AI-powered assistant to detect counterfeit medicines and provide trusted health guidance.

Introduction / Motivation :

Counterfeit medicines pose a serious threat to healthcare by reducing treatment effectiveness and endangering lives. Detecting them is difficult since counterfeiters closely mimic packaging and labeling. To tackle this, the project proposes an AI-powered assistant that detects counterfeit drugs using computer vision, provides authentic medical information, and educates users about risks and safe alternatives.

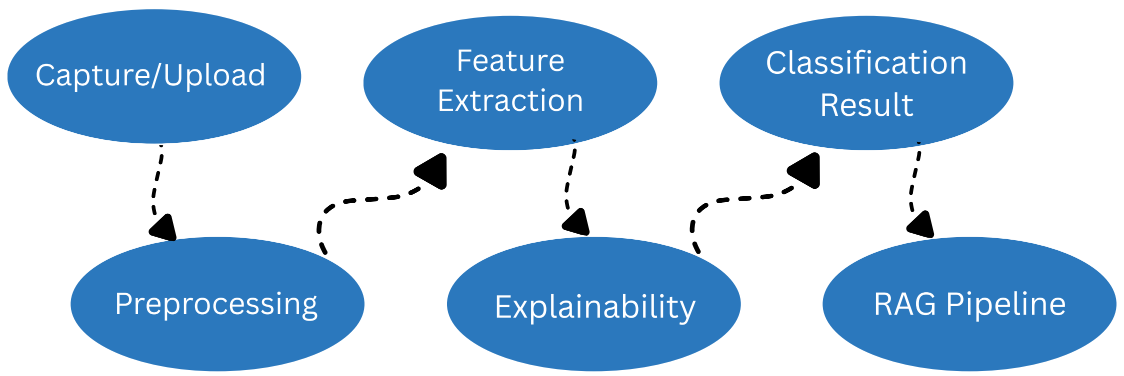
Dataset :

The dataset, sourced from Roboflow, consists of 1555 curated medicine images (packaging, strips, bottles, tablets) labeled as Authentic or Counterfeit. Metadata includes medicine names, and packaging types. Features focus on visual anomalies (font, color, hologram, misprints, alignment) and textual verification. The dataset is exported in YOLOv5 PyTorch format for training and deployment.

Roboflow: Counterfeit Med Detection

Literature Review :

Recent studies on counterfeit detection highlight the role of computer vision and AI-driven methods. Techniques like CNNs and Vision Transformers (ViTs) are widely applied for classifying authentic vs. counterfeit packaging, while Grad-CAM improves explainability by visualizing model decisions. In parallel, Retrieval-Augmented Generation (RAG) ensures reliable health information delivery from trusted sources such as WHO and FDA. Together, these works stress the need to combine AI-based detection with trustworthy medical knowledge.



Tasks :

The project involves collecting and preprocessing a dataset of authentic and counterfeit medicine images, ensuring they are suitable for model training. A Computer Vision model (CNN or Vision Transformer) will be trained to accurately classify medicines as authentic or counterfeit. To improve model transparency, Grad-CAM will be implemented to visualize which features influenced the model's decisions. Additionally, a RAG pipeline will be developed to retrieve reliable medical information from trusted sources like WHO and FDA. Finally, a chatbot assistant will be created to interact with users, providing guidance on medicine authenticity, associated health risks, and safe alternatives.

References :

[1] Perplexity
[2] Newton Faculty - Kartik Sir
[3] Kaggle,huggingface,Roboflow
[4] LangChain Documentation – RAG Pipelines.
[5] WHO – "Counterfeit medicines: Facts and Challenges."

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